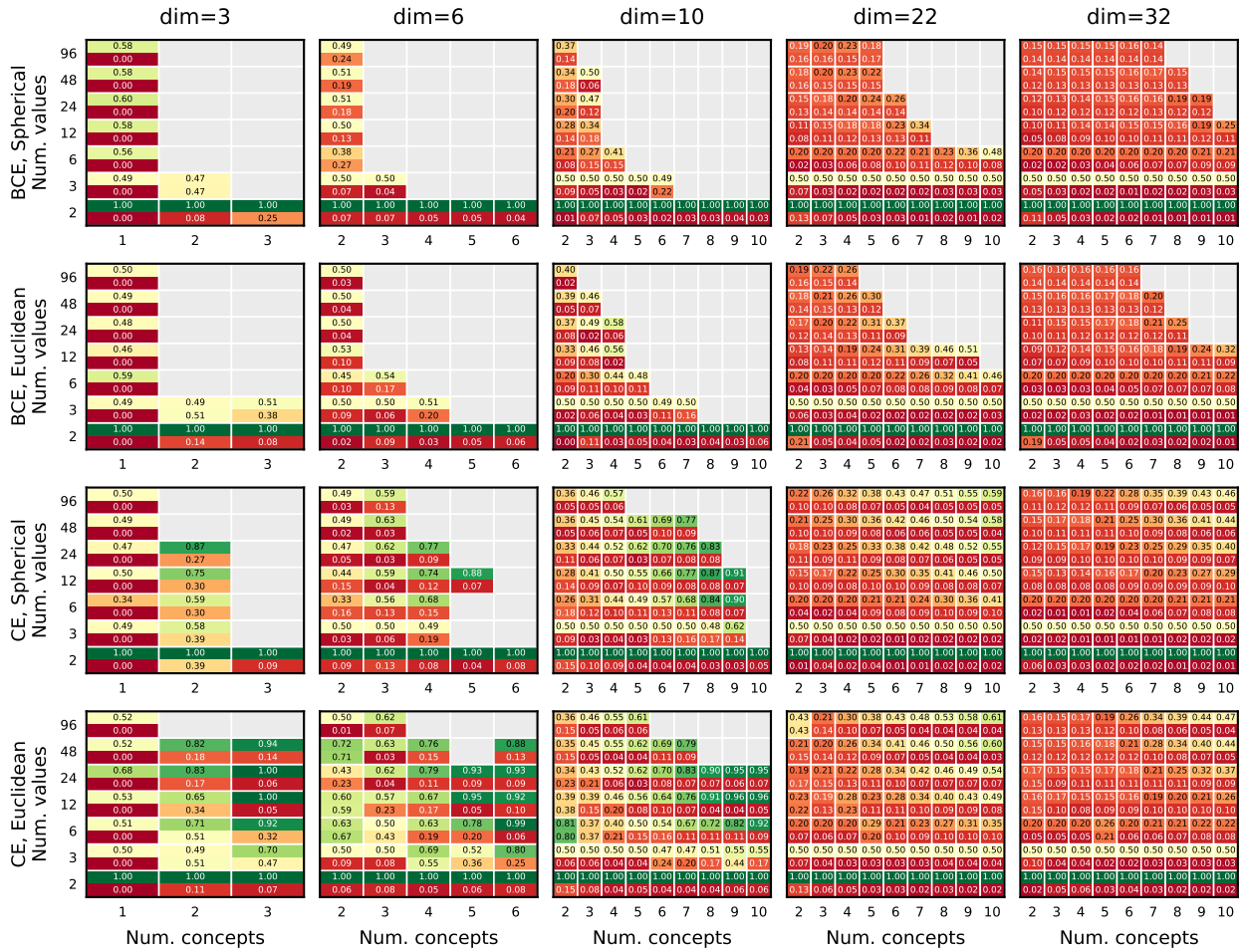

Figure 39. **Within-concept and cross-concept cosine similarity of factors in the from-scratch setting.** Metrics are computed as in Section I.2. For each (k, n) setting, each cell reports two values: within-concept cosine similarity (top) and cross-concept cosine similarity (bottom). As the number of concepts increases, cross-concept cosine similarity decreases, indicating stronger cross-concept orthogonality.

I.6.3. Results: Dimensionality.

In Section 4.3, we proved the lower bound $d \geq k$ for linear readout, independent of the number of values n. That result is a capacity bound: it states what is necessary in principle, but does not guarantee that a particular training objective reaches the bound.

Here, we estimate the minimum dimension needed to reach ≥ 0.99 per-concept accuracy across concept spaces varying in k and n, under CE (CLIP-like training) and BCE (SigLIP-like) losses, and under Euclidean (dot-product) and spherical (cosine-normalized) geometries.

The trends in Fig. 40 are consistent with the theory. CE (CLIP-like) in Euclidean space is typically close to the theoretical minimum ($d \approx k$), whereas BCE (SigLIP-like) generally requires larger dimension (often around $2k$), perhaps intuitively: in BCE each class is trained as an independent one-vs-rest binary problem, so each positive set must be separated from all negatives by its own hyperplane; CE instead uses relative margins across classes and can thus separate more flexibly (Crammer & Singer, 2002; Bangachev et al., 2025), matching the inference rule with the optimization objective. Spherical geometry shifts required dimensionality upward by roughly one dimension relative to Euclidean settings. Overall, this supports the theoretical claim that concept count is the primary dimensional driver, while showing that objective affect the attainability of the bound.

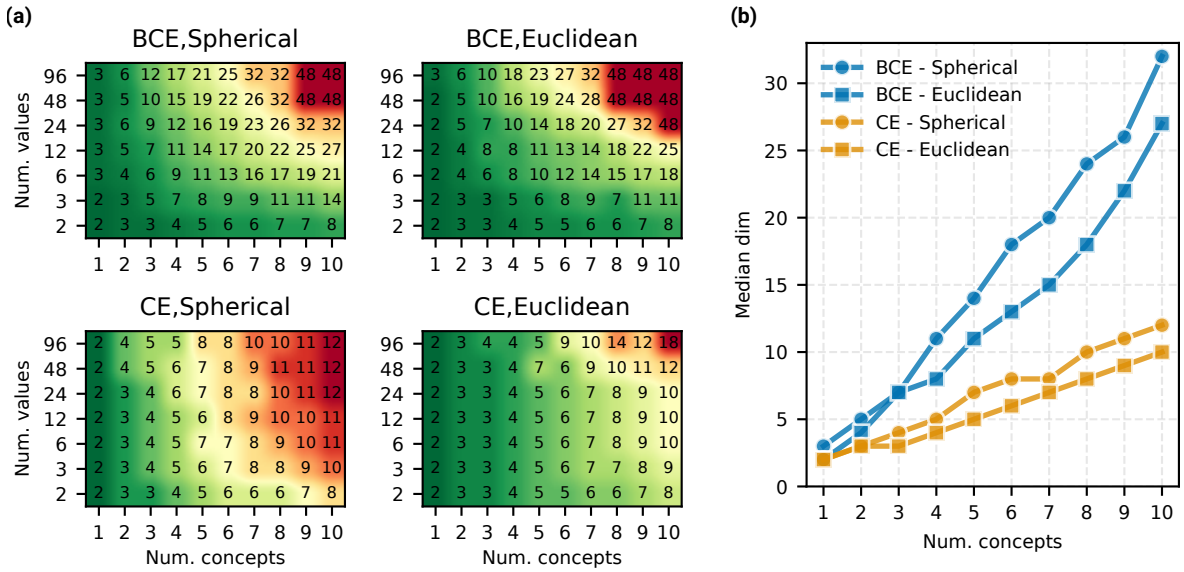

Figure 40. **Minimum embedding dimensionality across concept spaces, losses, and representation geometries.** We report the smallest dimension d needed to reach ≥ 0.99 per-concept classification accuracy for concept spaces with k concepts and n values per concept. **(a)** Required d as both k and n vary. **(b)** Median required d versus k: CE is typically near $d \approx k$, BCE needs roughly $2k$, and spherical variants require about one additional dimension compared to Euclidean.